Operating Systems

Processes

Chapter 3: Processes

- Process Concept
- Process Scheduling
- Operations on Processes
- Interprocess Communication

Process Concept (1/3)

Program vs. Process

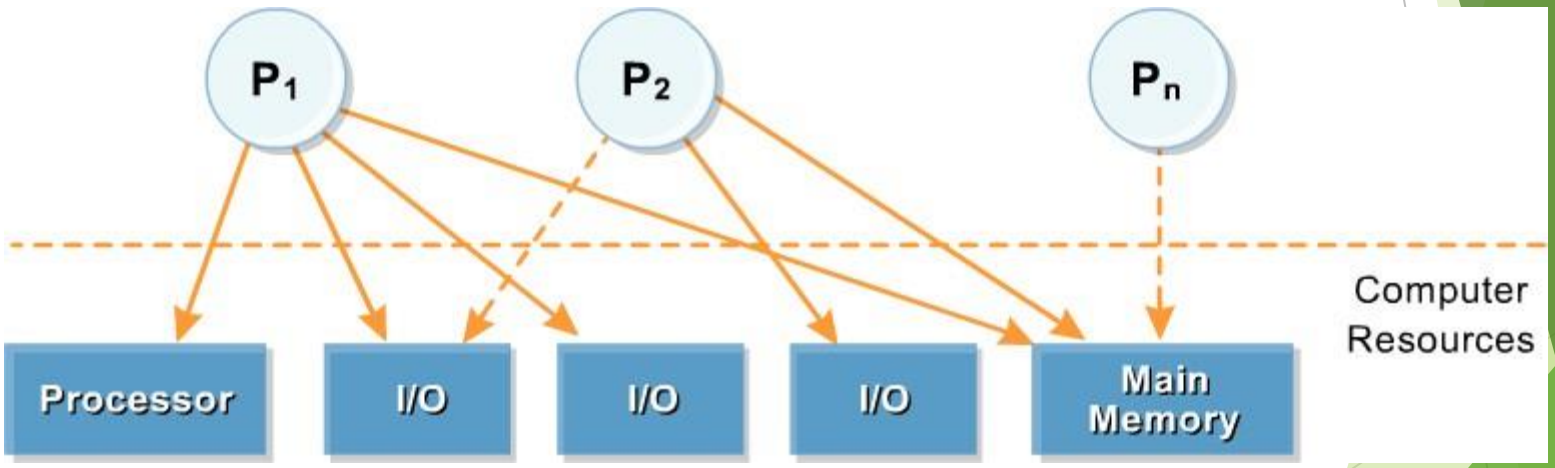
- **A program** is a **passive** entity such as the file that contains the list of instructions stored on a disk always referred to as an **executable file**.
- A program becomes a **process** when an executable file is loaded into the memory and then becomes an **active** entity.

Process Concept (2/3)

- The fundamental task of any operating system is the **process management**.
- Processes include not only a text but also include a set of resources such as open files and pending signals. Processes also contain internal kernel data, processor state, an address space, and a data section.

Process Concept (3/3)

- OS must **allocate resources to processes**, enable sharing of information, protect resources, and enable the synchronization among processes.



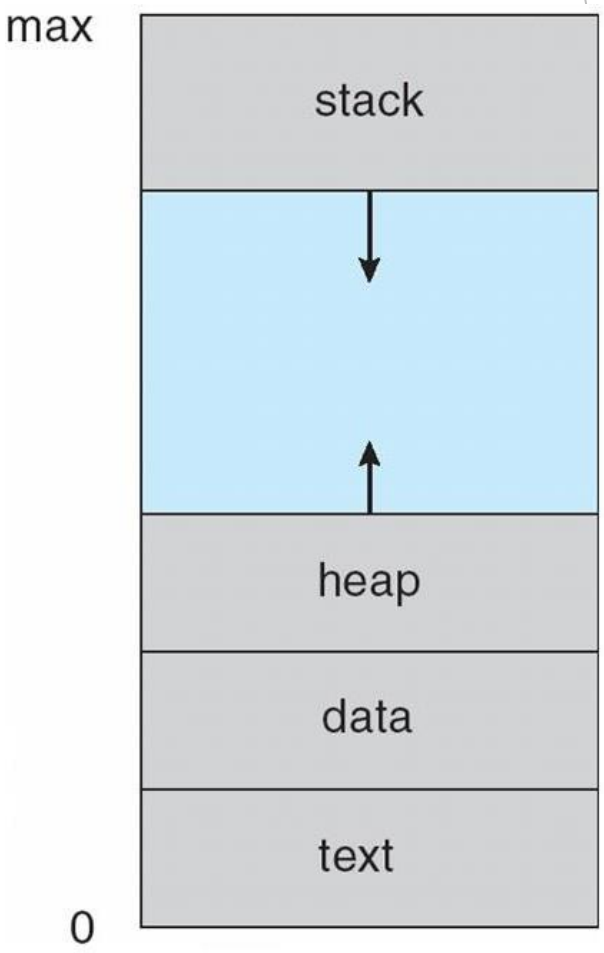
Process Elements (1/2)

- Segments of a process represents the following

  - components:
    - Text Section: the program code. This is typically read-only, and might be shared by a number of processes.
    - Data Section: containing global variables.
    - Heap: containing memory dynamically allocated during run time.
    - Stack: containing temporary data.
      - Function parameters, return addresses, local variables.

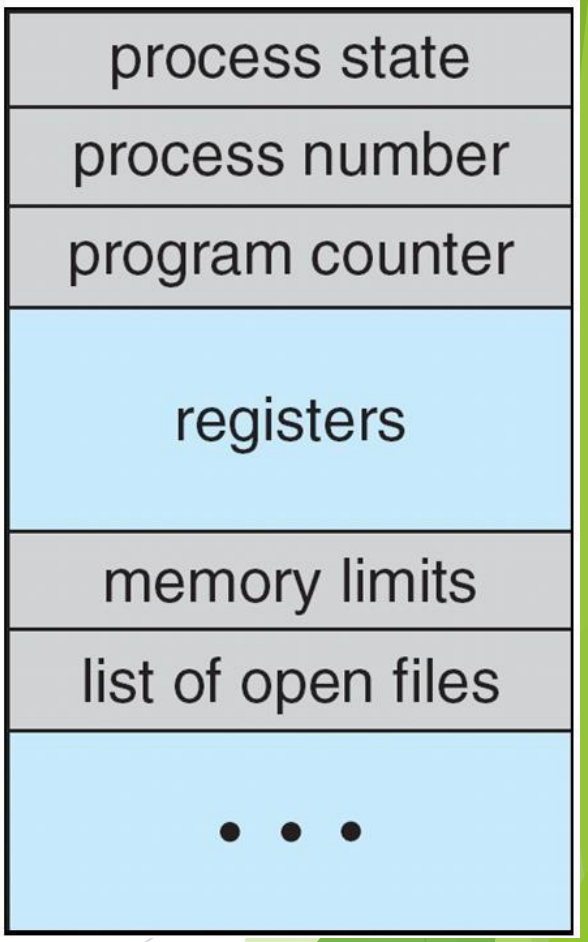
Process Elements (2/2)

- Process in Memory



Process Control Block (PCB)

- For better control of processes, operating systems need to consider their dynamic behaviors.
- Each process is represented in the OS by a Process Control Block (PCB).



Process Control Block (PCB) (2/2)

- Process Control Block (PCB) (1/3)
  - **Process identification information**
    - Process identifier: numeric identifiers represent the unique process identifier
    - User identifier: the user who is responsible for the job).
    - Identifier of the parent process that created this process.

Process Control Block (PCB) (2/2)

- Process Control Block (PCB) (2/3)
  - **Processor state Information**
    - Process state – running, waiting, etc
  - **Program counter**
    - location of instruction to next execute
  - **CPU registers**
    - contents of all process-centric registers

Process Control Block (PCB) (2/2)

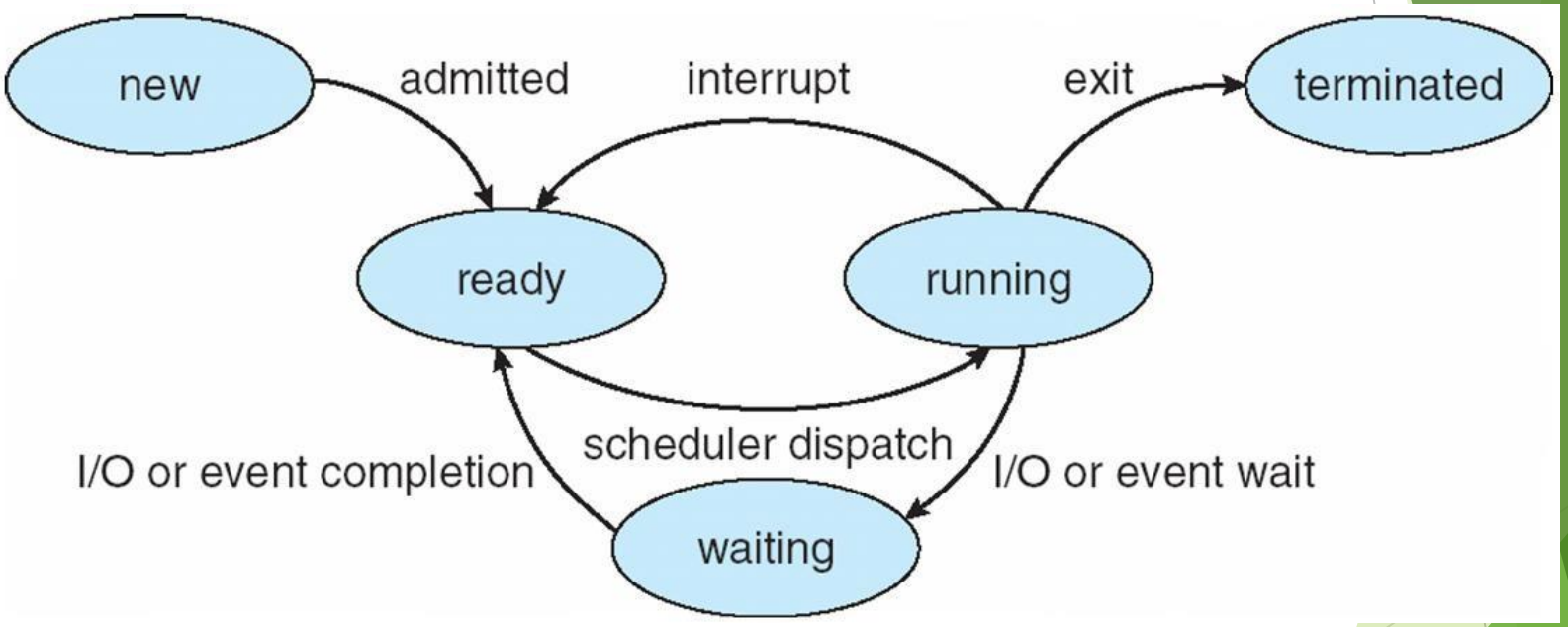
- Process Control Block (PCB) (3/3)
  - **CPU scheduling information**
    - priorities, scheduling queue pointers
  - **Memory-management information**
    - memory allocated to the process
  - **Accounting information**
    - CPU used, clock time elapsed since start, time limits
  - **I/O status information**
    - I/O devices allocated to process, list of open files

Process State (1/3)

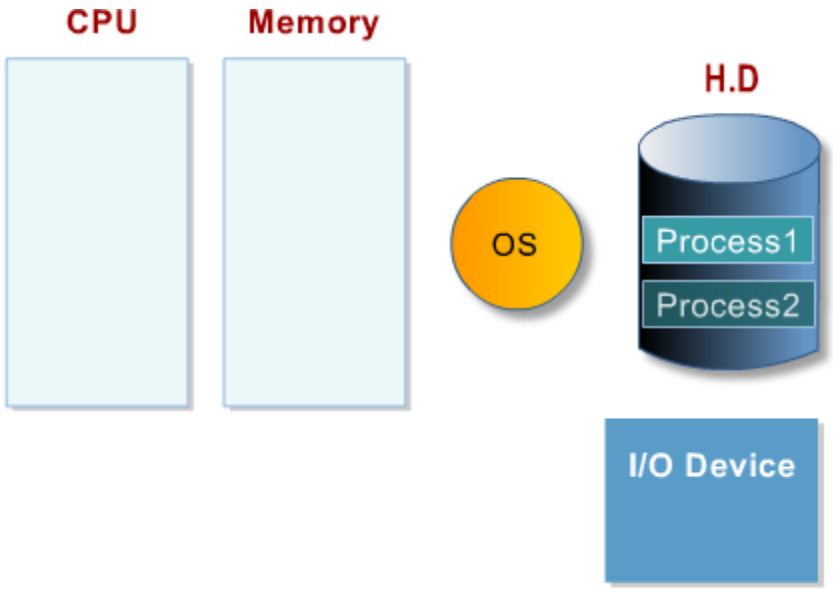
- As a process executes, it changes **state**
  - **new**: The process is being created
  - **running**: Instructions are being executed
  - **waiting**: The process is waiting for some event to occur
  - **ready**: The process is waiting to be assigned to a processor
  - **terminated**: The process has finished execution

Process State (2/3)

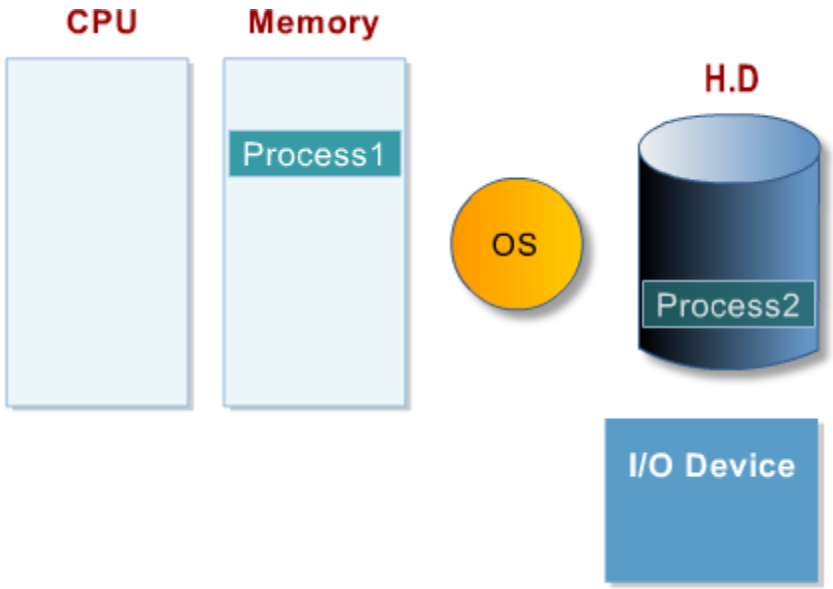
- Diagram of Process State



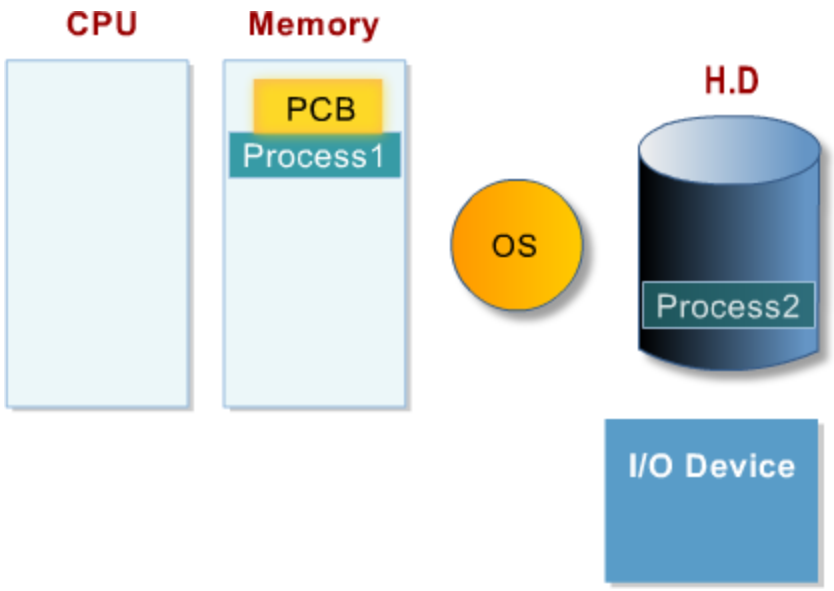
Process State (3/3)



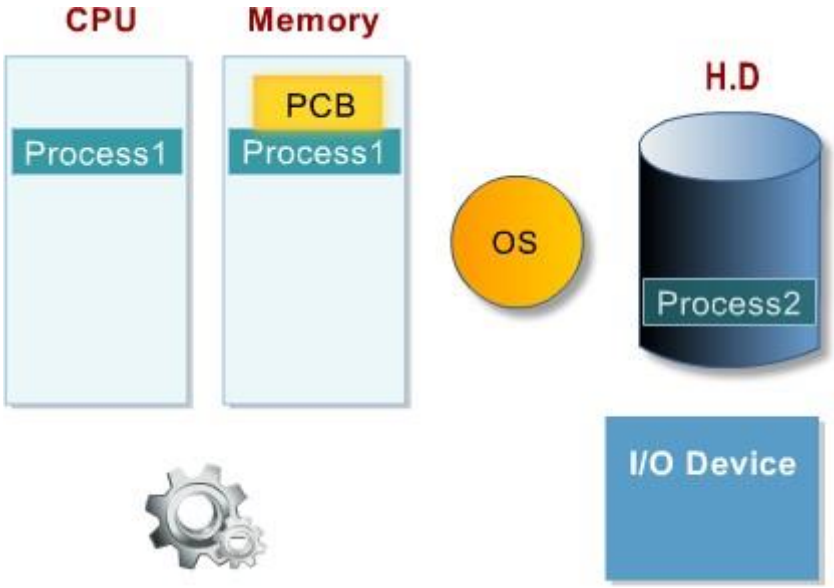
Process State (3/3)



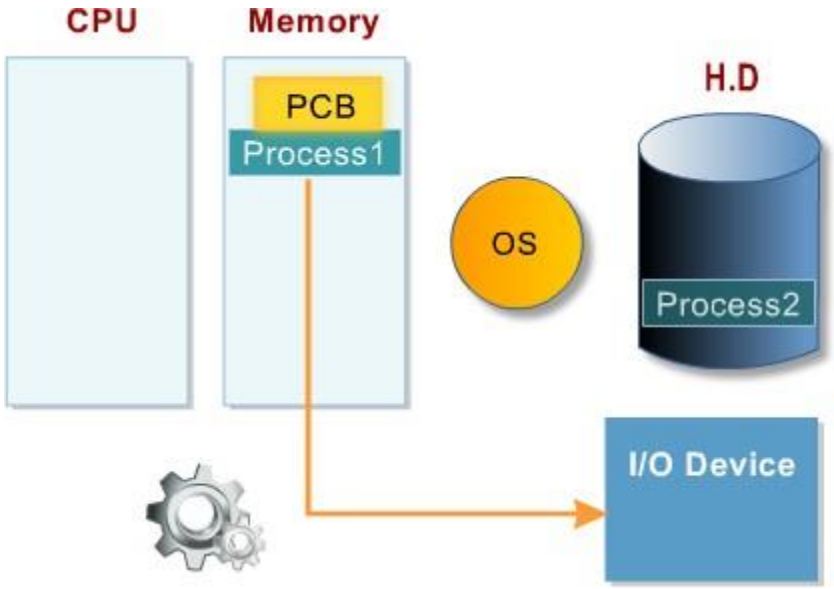
Process State (3/3)



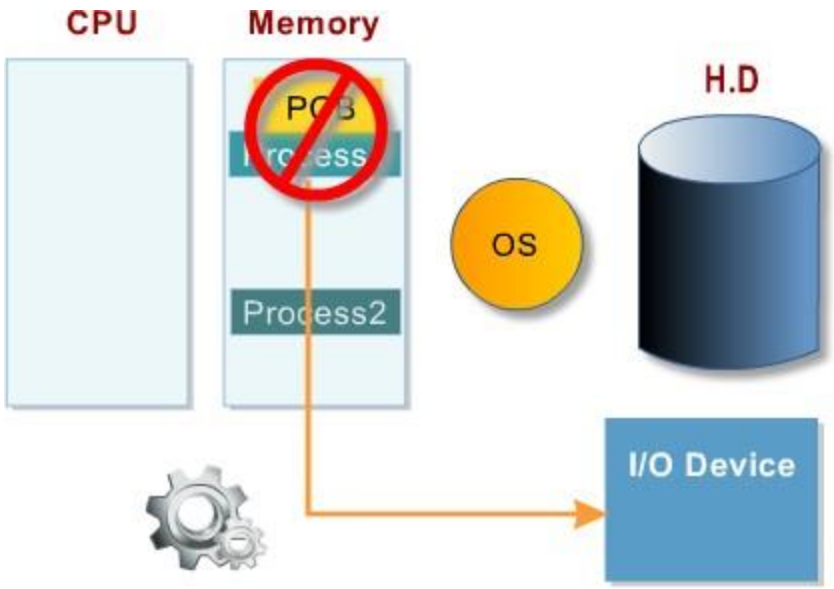
Process State (3/3)



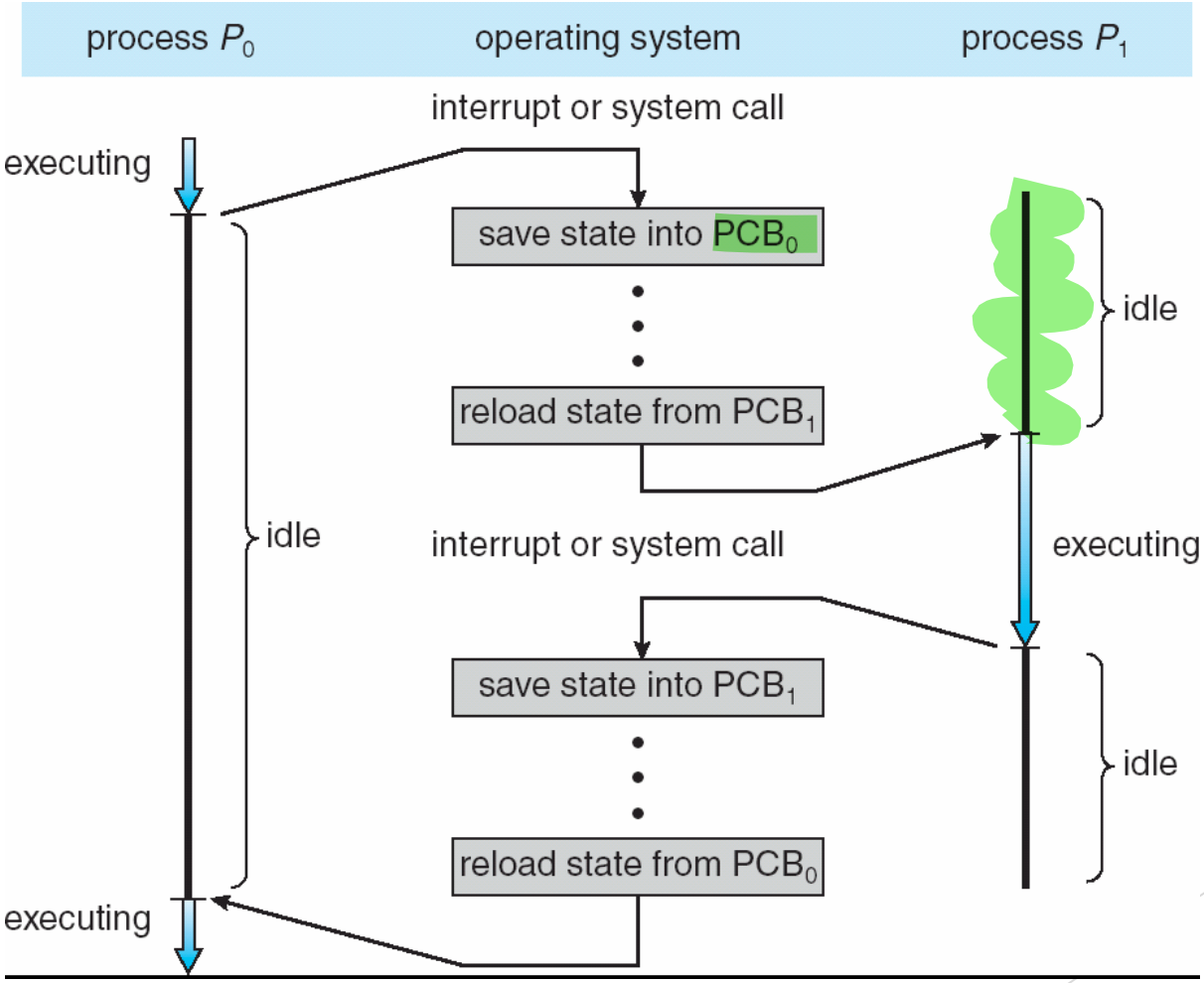
Process State (3/3)



Process State (3/3)

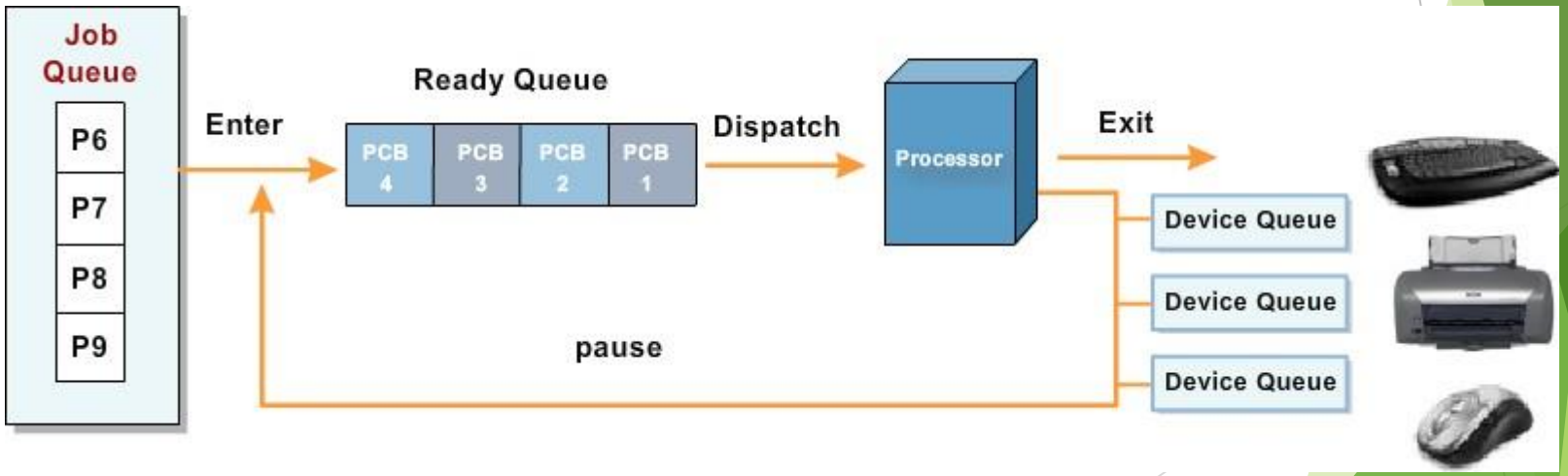


CPU Switch From Process to Process

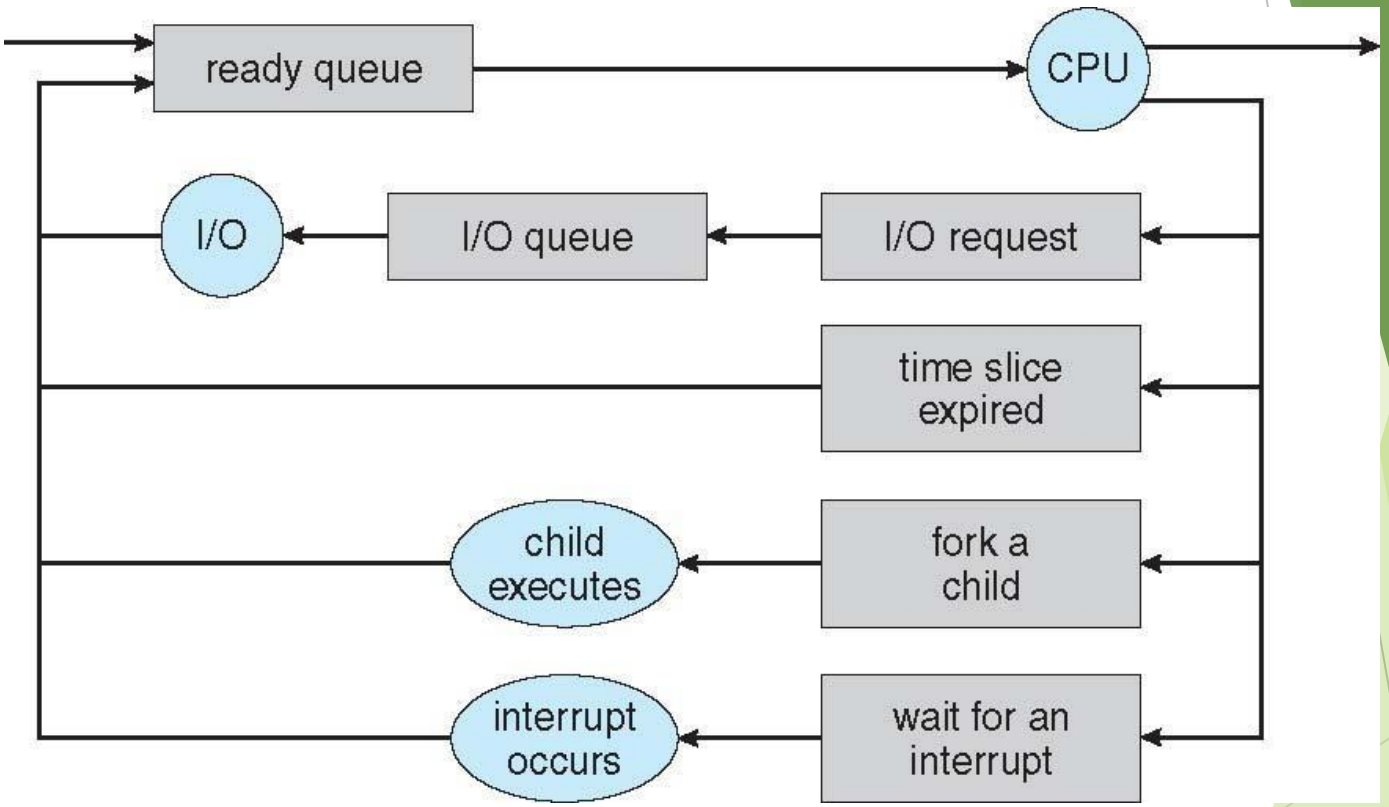


Process Scheduling (1/2)

- **Job queue** – set of all processes in the system
- **Ready queue** – set of all processes residing in main memory, ready and waiting to execute
- **Device queues** – set of processes waiting for an I/O device
- Processes migrate among the various queues



Process Scheduling (2/2)

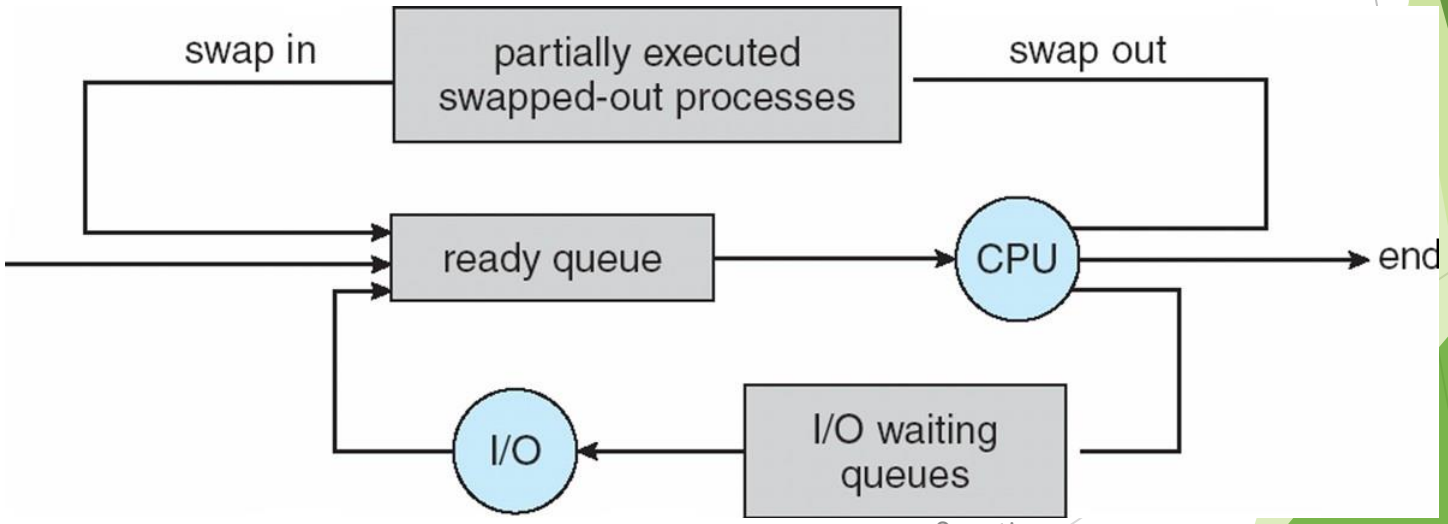


Schedulers (1/2)

- **Short-term scheduler (or CPU scheduler)**
  - Selects which process should be executed next and allocates CPU.
  - Invoked frequently (milliseconds) → (must be fast).
- **Long-term scheduler (or job scheduler)**
  - Selects which processes should be brought into the ready queue.
  - Invoked infrequently (seconds, minutes) → (may be slow).
  - Controls the degree of multiprogramming.

Schedulers (2/2)

- **Medium-term scheduler**
  - Can be added if degree of multiple programming needs to decrease
  - Remove process from memory, store on disk, bring back in from disk to continue execution: **swapping**



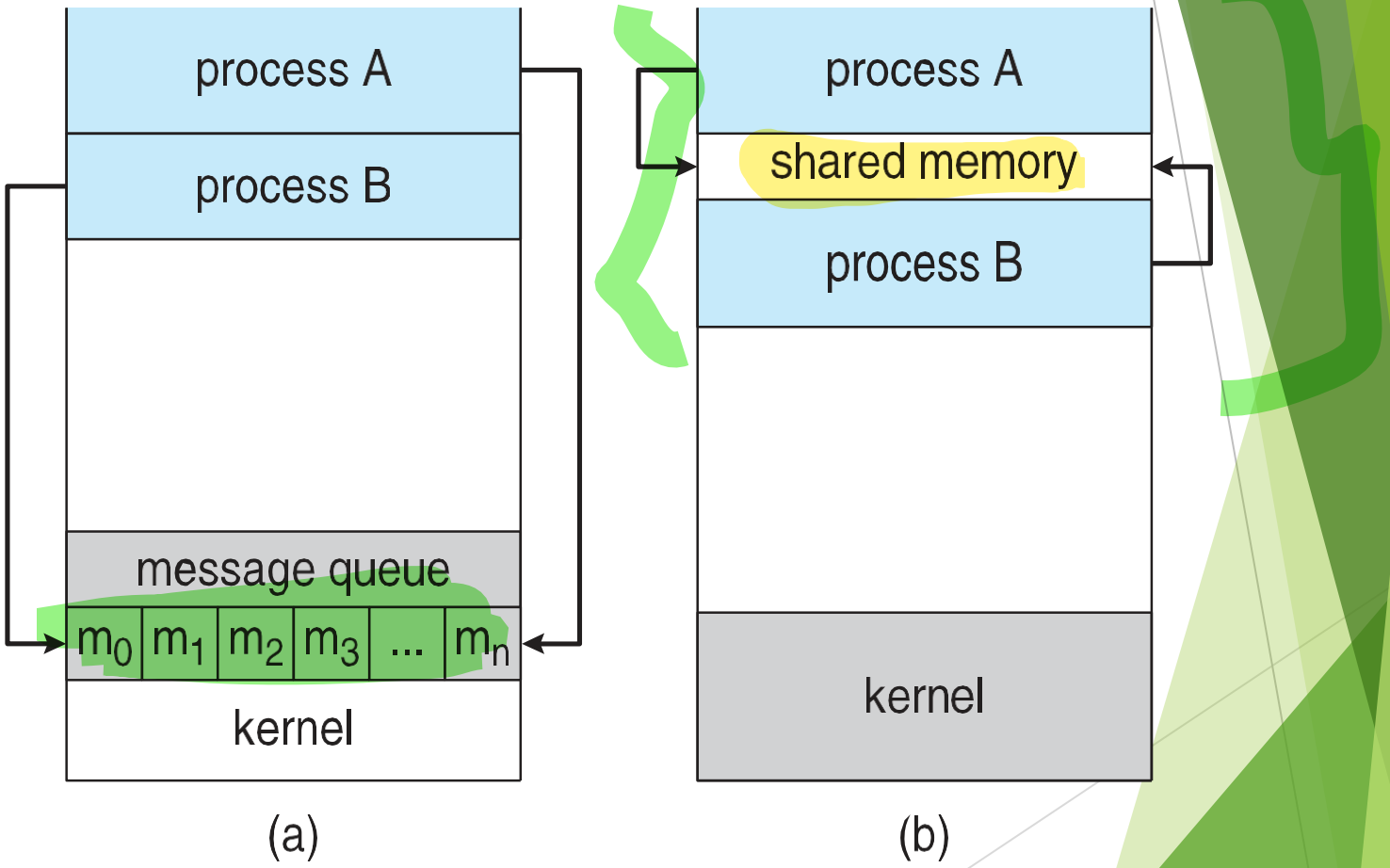
Interprocess Communication (1/2)

- Processes within a system may be **independent** or cooperating.
- Cooperating process can affect or be affected by other processes, including sharing data.
- Cooperating process need interprocess communication (IPC).

- <u>Two models of IPC:</u>
  - **Shared memory**
  - **Message passing**

Interprocess Communication (2/2)

(a) Message passing. **(b) shared memory.**



Thank You